

CSE3016 - AWS SOLUTION ARCHITECT

Presented By

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Module I

AWS Cloud Foundations: Introduction to Cloud Computing, Overview of AWS Global Infrastructure, AWS Shared Responsibility Model
AWS IAM - AWS Identity and Access Management (IAM), IAM Users, Groups, and Policies, IAM Roles, IAM Best Practices

Introduction to Cloud Computing

Introduction to Cloud Computing

- **Amazon Web Services (AWS)** is a platform that offers a wide range of **information technology services**.
- These services include **servers, storage, databases, networking, artificial intelligence, analytics**, and more.
- AWS enables **organizations and individuals** to access and use these services **flexibly and on demand**.
- It was **launched in 2006** by Amazon to help companies **reduce the cost and effort** of managing their own servers.
- Before AWS, businesses had to **buy and maintain physical computers and storage** for running websites and applications.
- This traditional setup required **significant time, money, and technical effort**.
- AWS solved this problem by providing **computing resources over the internet (cloud)**.
- As a result, users can **scale their infrastructure easily** without investing in physical hardware.

Before AWS

01



High Upfront
Infrastructure
Costs

02



Limited
Scalability

03



Long Time
Deployment

05



Global Reach
Was Difficult

04



Complex Disaster
Recovery

After AWS



01

No Upfront
Costs



02

Instant
Scalability



03

Faster
Deployment



05

Global Reach
with a Few Clicks



04

Built-in Disaster
Recovery & High
Availability

How AWS Delivers Cloud Services

- **Amazon Web Services (AWS)** delivers cloud computing through a **global network of data centers** connected by **high-speed fiber networks**.
- These data centers host a variety of services such as **storage, computing power, databases, and machine learning tools**.
- Users can **access these services on demand via the internet**, without owning physical infrastructure.
- AWS enables **startups and enterprises** to **build and run applications** without managing hardware.

- AWS takes care of essential operations like:
 - *Server maintenance*
 - *Security patching*
 - *Resource scaling*
 - *High availability*
- For example, **Amazon RDS (Relational Database Service)** automates backend tasks such as **database setup, backups, and failover**.
- This allows users to **launch and manage databases quickly** — often within minutes and with minimal effort.

Key Features and Services of AWS

1. Elastic Infrastructure:

AWS provides the ability to **scale resources up or down** automatically based on changing demand.

This is especially beneficial for **websites or applications with fluctuating traffic**.

2. Wide Service Portfolio:

AWS offers a **broad range of services** including **computing, storage, databases, artificial intelligence, analytics, networking, security, and IoT**.

This allows users to choose and combine services as needed for different applications.

3. Scalability:

AWS is designed to **support large-scale projects and enterprise-level applications**.

Users can easily **expand or reduce resources** to meet workload requirements efficiently.

4. Flexible Payment Model:

AWS follows a **pay-as-you-go** pricing model, meaning users **pay only for the resources they actually use**.

This helps organizations **optimize costs** and avoid unnecessary expenses.

How is AWS able to offer pay-as-you-go?

- **AWS uses virtualization technology** to separate multiple **virtual machines (VMs)** running on the same physical hardware.
- This means **different companies can share the same physical server**, but their virtual environments remain **secure and isolated**.
- By **maximizing the utilization of server resources**, AWS ensures that its hardware is used efficiently.
- This efficient resource sharing allows AWS to **reduce operational costs**.
- As a result, AWS can offer customers a **cost-effective, pay-as-you-go pricing model**.
- Users only pay for the **resources they actually use**, instead of maintaining expensive, dedicated servers.



**Pay-as-You-Go
Pricing**



**On-Demand
Instances**



**Reserved
Pricing**



**Spot
Instances**

AWS Pricing



Free Tier



**AWS Pricing
Calculator**



**Cost Management
and Optimization**

5. Rapid Deployment:

AWS enables **quick deployment and updates** of applications and services, helping businesses launch faster.

6. High Security:

AWS implements **strong security measures and specialized services** to ensure the **protection of user data**.

7. Global Infrastructure:

AWS operates **numerous data centers across the world**, providing **high availability, reliability, and low latency** for users everywhere.

8. Ease of Management:

AWS offers a **user-friendly management console** that allows users to **easily monitor, manage, and configure resources**.

- **Comprehensive Cloud Platform:**

Amazon Cloud Services serves as a **complete platform** that supports **data storage, computing power, application deployment, and database management** for **businesses, developers, and individuals** of all sizes.

Real-World Use Cases of AWS Services

- **Netflix:**

- Uses **AWS for storage and application scaling** to manage massive user demand.
- Ensures **seamless content delivery** with **low latency and no interruptions** for millions of users worldwide.

- **Airbnb:**

- Utilizes **AWS to manage workloads** efficiently across its global operations.
- Provides a **scalable and reliable infrastructure** to support its **virtual marketplace and lodging services**.

- **NASA's Jet Propulsion Laboratory (JPL):**

- Leverages **AWS to process and analyze large volumes of scientific data**.
- Supports **research missions and space exploration** through **high-performance computing and data management** capabilities.

Core AWS Compute Services



Amazon
Cloudfront



Lambda: Run Code
Without Servers



Amazon
SNS



AWS
AutoScaling



EC2 Virtual Servers
You Control



Lambda: Run Code
Without Servers



Containers: ECS, EKS
for Scalable Apps

Cloud Computing

- **Computing Paradigm:**

Cloud computing is a **modern computing model** that provides **on-demand access** to computing resources over the internet.

- **Shared Pool of Resources:**

It offers a **shared pool of configurable resources**, including **servers, storage, databases, networking, software, and analytics**.

- **On-Demand Provisioning:**

Users can **provision and release resources quickly** with **minimal management effort** or **interaction with the service provider**.

- **Scalability and Cost-Effectiveness:**

Cloud computing is **highly scalable** and **cost-efficient**, making it suitable for organizations of all sizes.

- **Definition:**

It refers to the delivery of computing services via the internet, where users are billed based on usage, similar to utilities like water or electricity.

Client-Server Model

- The **client-server model** is an important concept in **cloud computing**.
- It involves **many clients** accessing services from a **centralized server**.
- The **client** sends **requests** to the **server**, which processes and returns the results.
- The **client** is the interface through which a **user interacts** with the **server**.
- The **server** provides the **resources or services** requested by the client.

Client



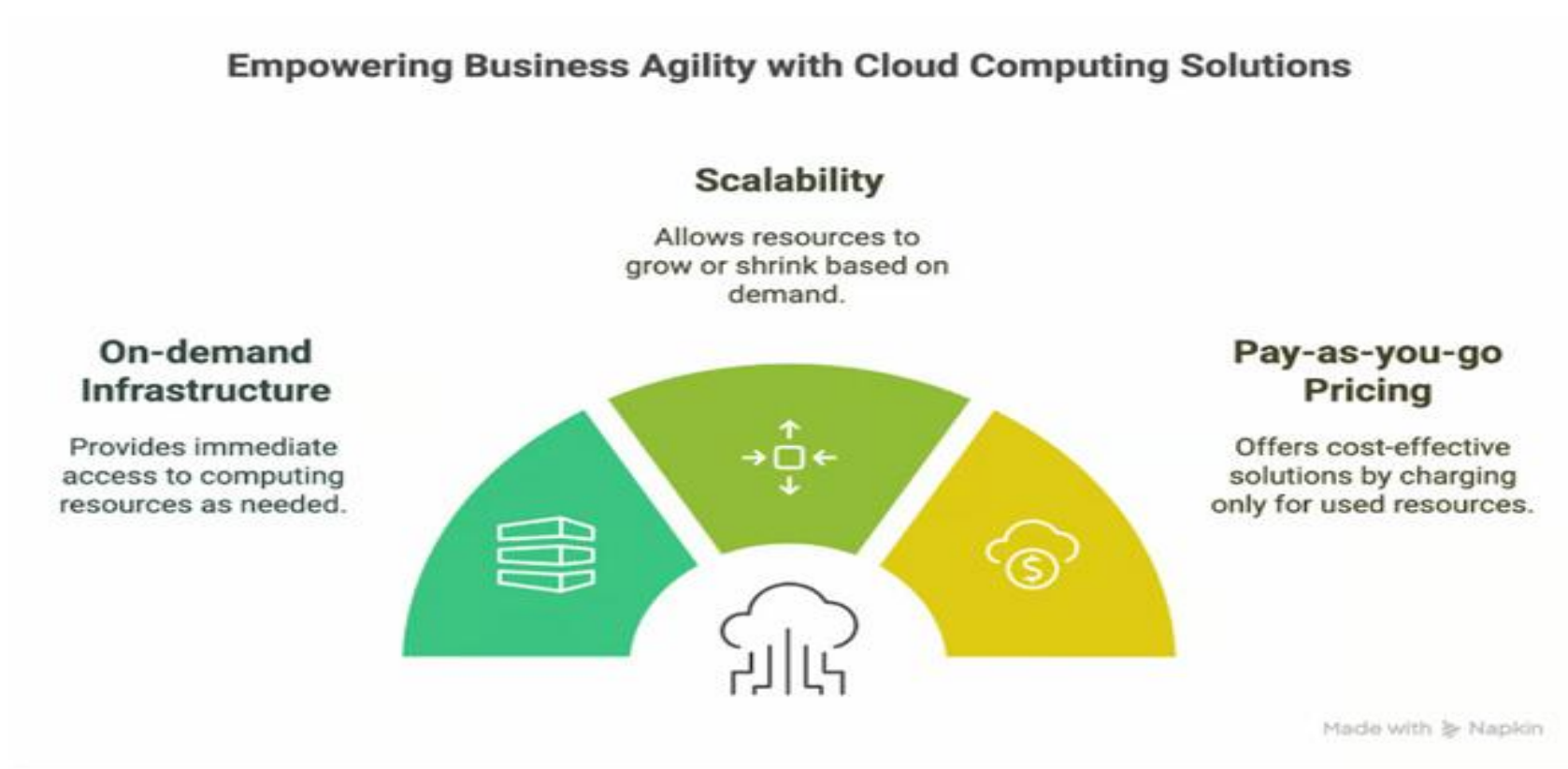
Server



Delivery Models

1. Infrastructure as a Service (IaaS):

Virtualized computing resources online.



A diagram showing the key features of Infrastructure as a Service (IaaS).

Features of IaaS (Infrastructure as a Service):

- **On-Demand Infrastructure:**

Users can **scale resources** such as storage, computing power, and bandwidth **as needed**, without buying extra hardware.

- **Scalability:**

IaaS offers **flexible scaling options**, allowing businesses to **increase or decrease resources** based on current workload or demand.

- **Pay-as-You-Go Pricing:**

IaaS uses a **pay-per-use model**, meaning users **only pay for the resources they actually use**, helping reduce operational costs.

Common use cases of IaaS

- **Hosting Websites:**

IaaS is widely used for **website hosting**, providing the necessary **computing power and network capacity** to manage large volumes of traffic.

- **Disaster Recovery:**

Businesses use IaaS for **backup and disaster recovery**, ensuring **high availability and data protection** without heavy investment in physical servers.

- **Test Environments:**

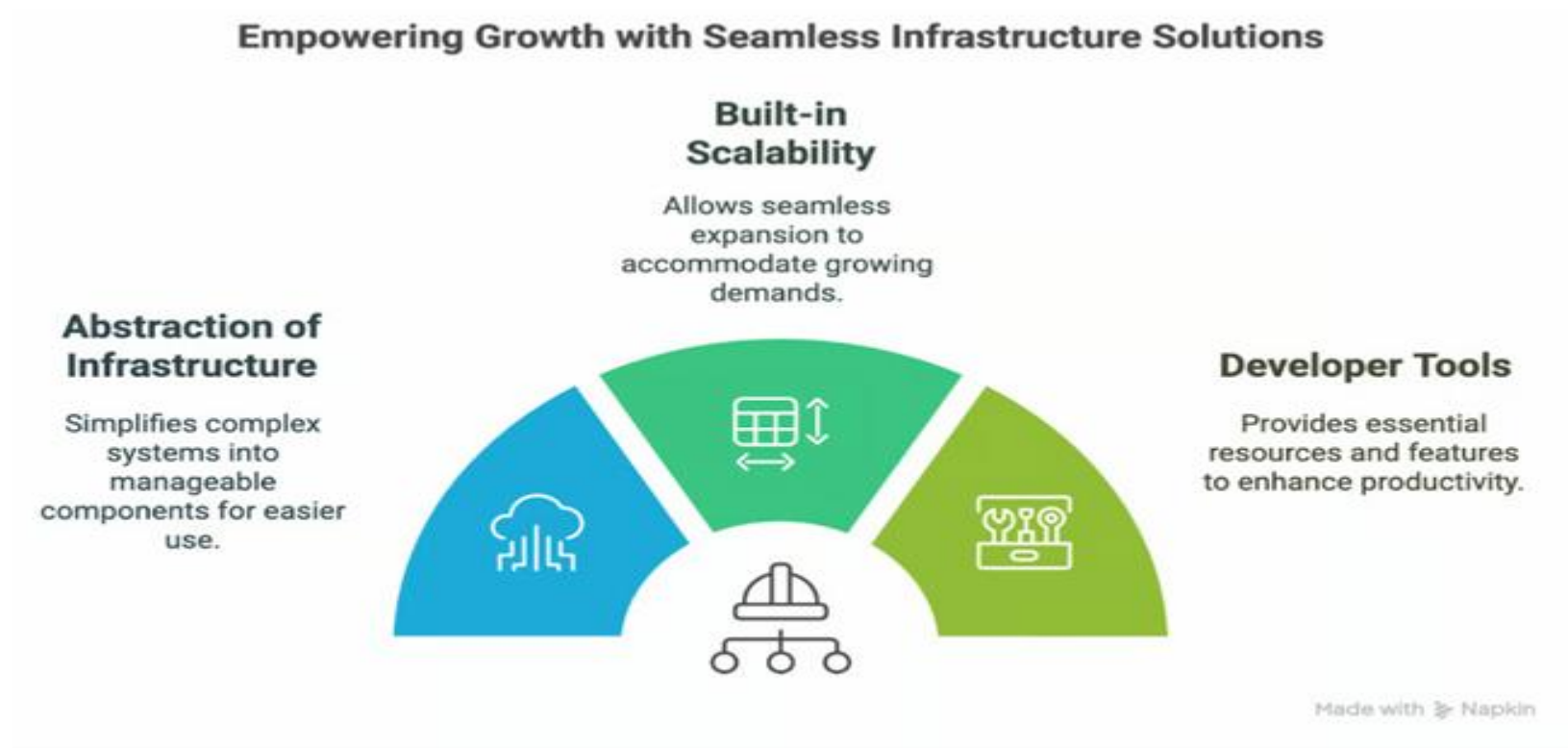
Developers use IaaS to **create virtual test environments** where they can **quickly deploy and test applications** without affecting live production systems.

Popular IaaS providers

- **AWS EC2 (Amazon Elastic Compute Cloud):**
A leading IaaS service by Amazon Web Services, offering **scalable and flexible compute capacity** in the cloud.
- **Google Compute Engine:**
Google's IaaS platform that provides **virtual machines and cloud storage** for running applications and workloads.
- **Microsoft Azure VMs:**
Microsoft Azure's **virtual machine service** that allows users to **run virtual servers and scale resources dynamically** based on demand.

2. Platform as a Service (PaaS):

Cloud-based app development platforms.



A diagram showing the key features of Platform as a Service (PaaS).

Features of PaaS (Platform as a Service):

- **Abstraction of Infrastructure:**

PaaS hides the complexity of **server and storage management**, allowing developers to **focus on building and deploying applications**.

- **Built-in Scalability:**

PaaS platforms provide **automatic scaling**, enabling applications to **handle increased demand** without manual intervention.

- **Developer Tools:**

PaaS includes **integrated development tools**, such as **APIs, databases, and frameworks**, which help **simplify and speed up** the application development process.

Common use cases of PaaS (Platform as a Service):

- **Web App Development:**

PaaS is widely used for **developing and deploying web applications** while minimizing the need to manage underlying infrastructure.

- **API Management:**

Developers use PaaS to **create, deploy, and manage APIs**, enabling **smooth communication** between different applications and services.

- **Microservices Architecture:**

PaaS supports the development of **microservices**, which are **independent, scalable components** that can be deployed and managed separately.

Popular PaaS (Platform as a Service) providers:

- **Heroku:**

A widely used PaaS platform that enables developers to **build, run, and deploy web applications** easily **without managing infrastructure**.

- **Google App Engine:**

Google's PaaS service that offers a **fully managed environment** for building and deploying applications in multiple languages such as **Python, Java, and Go**.

- **AWS Elastic Beanstalk:**

Amazon's PaaS solution that allows developers to **deploy and manage applications automatically**, taking care of **server provisioning and scaling** behind the scenes.

3. Software as a Service (SaaS)

Web-based applications accessible via browsers.



A diagram showing the key features of Software as a Service (SaaS).

Features of SaaS (Software as a Service):

- **Fully Managed by the Vendor:**

The **service provider handles all updates, security patches, and maintenance**, reducing the management burden on users.

- **Subscription-Based Pricing:**

SaaS typically follows a **subscription model**, allowing businesses to **pay monthly or annually**, making it **affordable and flexible**.

- **Browser-Based Access:**

SaaS applications are **accessible through web browsers**, eliminating the need for **local installations or specific hardware**.

Common use cases of SaaS (Software as a Service):

- **Email Services:**

SaaS applications like **Gmail** and **Outlook** provide **cloud-based email communication**.

- **Customer Relationship Management (CRM):**

SaaS platforms such as **Salesforce** help businesses **manage customer relationships, track sales, and analyze data**.

- **Project Management:**

Tools like **Trello** and **Asana** offer **collaborative platforms for task and project management**.

Popular SaaS examples:

- **Salesforce:**

A leading **CRM SaaS platform** that helps businesses **manage customer relationships and sales pipelines**.

- **Google Workspace:**

A **cloud-based productivity suite** including **Docs, Sheets, Drive, and more**, delivered as SaaS.

- **Microsoft 365:**

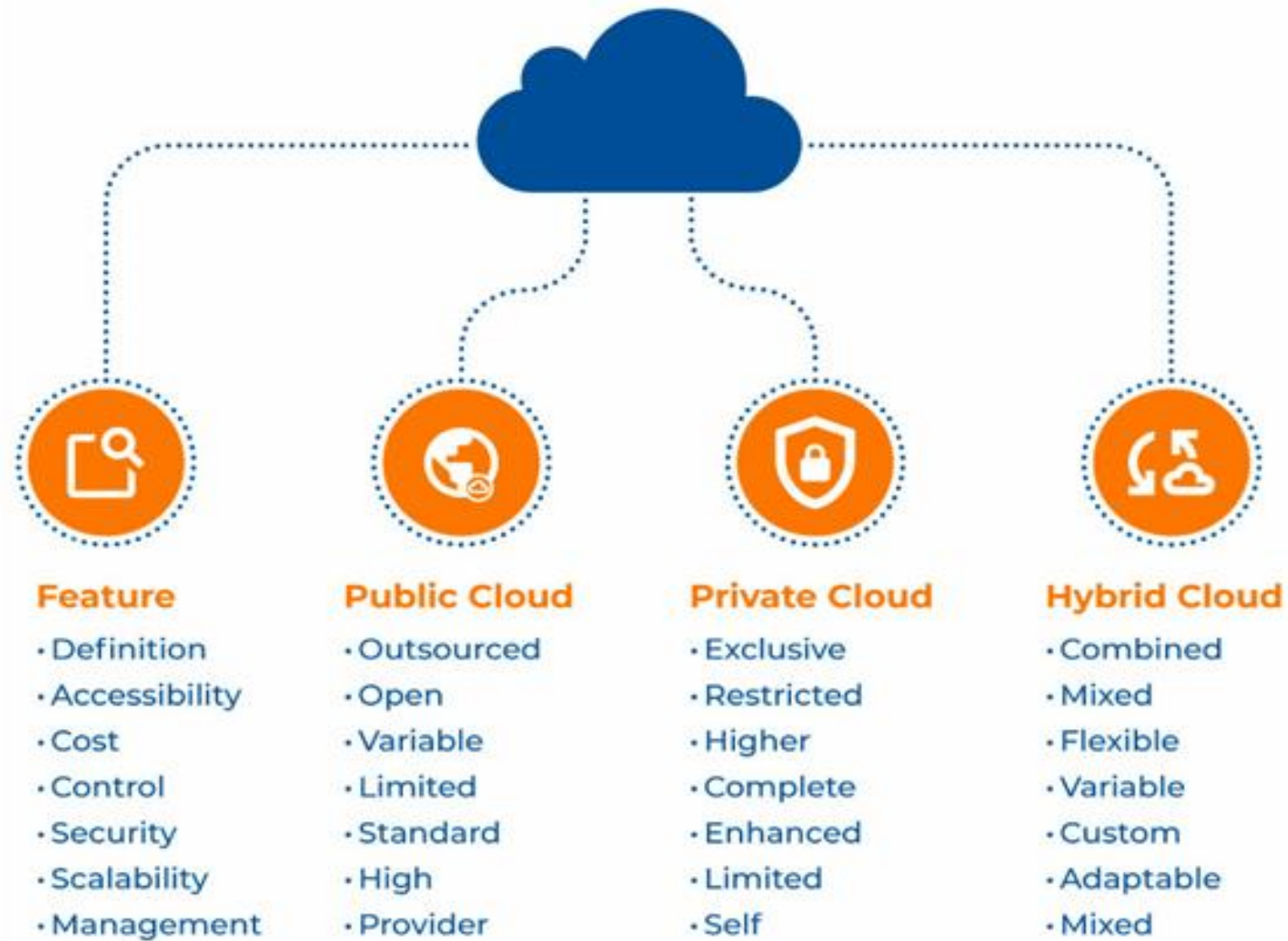
Microsoft's **cloud-based suite** for **word processing, spreadsheets, email, and collaboration**, accessible from anywhere via the internet.

Comparison of IaaS, PaaS and SaaS

	IaaS (Infrastructure as a Service)	PaaS (Platform as a Service)	SaaS (Software as a Service)
User Access/Identity	Customer Responsibility	Customer Responsibility	Customer Responsibility
Data	Customer Responsibility	Customer Responsibility	Customer Responsibility
Application	Customer Responsibility	Customer Responsibility	Cloud Service Provider Responsibility
Guest OS	Customer Responsibility	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility
Virtualization	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility
Network	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility
Infrastructure	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility
Physical	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility	Cloud Service Provider Responsibility

Deployment Models

- **Public cloud:** Accessible via the internet, provided by third parties.
- **Private cloud:** Exclusively operated for a single organization.
- **Hybrid cloud:** Mix of private and public cloud resources.
- *(Refer -> <https://aws.amazon.com/what-is/private-cloud/>)*
- *(Refer -> <https://www.geeksforgeeks.org/computer-science-fundamentals/cloud-deployment-models/>)*



Overview of AWS Global Infrastructure

- **Largest Cloud Infrastructure:**
AWS operates the **world's largest cloud computing infrastructure**, providing **reliable and scalable services globally**.
- **Global Network of Data Centers:**
The infrastructure consists of **data centers and resources** strategically located worldwide to **deliver cloud services efficiently**.
- **Components of AWS Infrastructure:**
Key components include **Regions, Availability Zones (AZs), Local Zones, and Wavelength Zones**.
- **Availability Zones (AZs):**
AZs are **isolated data center locations within AWS Regions** and form the main building blocks of the infrastructure.
- **Current Coverage:**
AWS has **102 Availability Zones across 32 geographic regions globally**.
- **Future Expansion:**
Plans include **12 additional Availability Zones and 4 new Regions** in countries such as **Canada, Malaysia, New Zealand, and Thailand**.

Local Zones:

- AWS provides **35 Local Zones** that extend services to **select metropolitan areas**, bringing resources closer to **end-users** and **reducing latency**.

Wavelength Zones:

- AWS has **29 Wavelength Zones**, developed in collaboration with **telecommunication providers** to place services near **5G networks**.

Purpose:

Both Local and Wavelength Zones are designed to **support ultra-low latency applications**, improving **performance for latency-sensitive workloads**.



AWS Global Infrastructure



Every region is **connected** via a high bandwidth, fully redundant network

There are 25 **regions** around the world

A **Region** is a physical location in the world and is **independent**

Each region consists of **two** or **more** Availability Zones

An Availability Zone is composed of **one** or **more** data centers



Local Zones extend regions closer to end-users



AWS Local Zones



1. Core components of AWS Global Infrastructure:

- **Regions:**

Physical locations around the world where AWS **clusters data centers**. Each region is **independent** and contains multiple **Availability Zones**.

- **Availability Zones (AZs):**

Groups of **discrete data centers** within a region, featuring **redundant power, networking, and connectivity** to ensure **fault tolerance and high availability**.

- **Edge Locations:**

Over **400 sites** used by services like **Amazon CloudFront** to deliver **content globally with low latency**.

- **Local Zones & Wavelength Zones:**

Provide **compute and storage closer to users** or within **telecom networks**, supporting **ultra-low latency applications** such as **gaming or real-time analytics**.

- **Outposts:**

Extend AWS infrastructure **on-premises**, enabling **hybrid cloud solutions**.

2. AWS Regions:

- **Definition:**

An **AWS Region** is a distinct **geographical area** worldwide where a **collection of data centers** is clustered.

- **Availability Zones in Regions:**

Each Region contains **multiple Availability Zones (AZs)**, with at least **three AZs per Region** for **geographic redundancy** and **high availability**.

- **Difference from AZs:**

- **Regions** are **larger geographical areas**.
- **Availability Zones** are **subsets of Regions**, consisting of individual data center groups.

- **Global Presence:**

AWS operates in multiple regions across **North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East**.

3. AWS Availability Zones (AZs):

- **Definition:**

Availability Zones (AZs) are **physically separated data centers** within an AWS Region.

- **Independent Infrastructure:**

Each AZ has its own **power, cooling, and networking infrastructure**.

- **Purpose:**

AZs are designed to provide **redundancy and resiliency** for applications and services.

- **High Availability:**

Deploying resources across **multiple AZs** ensures that if one AZ experiences **issues or downtime**, others **continue operating**, enhancing the **overall resilience** of AWS-hosted applications.

- **Importance of Physical Separation:**
Physical distance between an **AWS Region** and its **AZs** helps **minimize the impact of natural disasters or localized incidents**.
- **Substantial Separation:**
Availability Zones within a region are **significantly separated** to ensure **resiliency**, yet remain **relatively close** for low-latency communication.
- **Distance Limit:**
All AZs within a given AWS Region are typically **within 100 kilometers (≈60 miles)** of each other.
- **Global Coverage:**
AWS maintains **multiple data center locations across all regions** to provide **high availability and fault tolerance**.

Europe / Middle East / Africa

- 11 Geographic Regions
- 33 Availability Zones
- 39 Edge Network locations
- 2 Regional Edge Cache locations

Availability Zones:

- Bahrain (3), Cape Town (3), Frankfurt (3), Ireland (3), London (3), Milan (3), Paris (3), Spain (3), Stockholm (3), Zurich (3), and UAE (3)

North America

- 7 Geographic Regions
- 25 Availability Zones
- 44 Edge Network locations
- 2 Regional Edge Cache locations

Availability Zones:

N. Virginia (6), Ohio (3), N. California (3), Oregon (4), US-East (3), US-West (3), Central (3) South America

- 1 Geographic Regions
- 3 Availability Zones (in São Paulo)
- 4 Edge Network locations
- 1 Regional Edge Cache locations

Asia Pacific and China

- 12 Geographic Regions
- 38 Availability Zones
- 34 Edge Network locations
- 5 Regional Edge Cache locations

4. AWS Local Zones:

- **Definition:**

AWS Local Zones are **part of the AWS Global Infrastructure** that bring **cloud services closer to specific metropolitan areas** with a high concentration of users and applications.

- **Relationship with AZs:**

Local Zones are a **subset of AWS Availability Zones**, but located **near end-users** to reduce latency.

- **Purpose:**

They enable customers to **deploy applications requiring low latency** to end-users or specific resources in that area.

- **Use Cases:**

Particularly useful for **real-time processing applications**, such as **gaming, interactive multimedia, and financial services**, where **low latency is critical** for a seamless user experience.

5. AWS Local Zones Locations:

- **Global Presence:**

AWS Local Zones are available in **33 metropolitan areas worldwide**.

- **US Locations:**

16 Local Zones are located within the **United States**.

- **International Locations:**

17 Local Zones are located **outside the US** (specific locations not listed in the text).

1. Lagos
2. Auckland
3. Santiago
4. Bangkok
5. Kolkata
6. Copenhagen
7. Hamburg
8. Helsinki
9. Lima
10. Muscat
11. Perth
12. Querétaro
13. Taipei
14. Warsaw
15. Delhi
16. Buenos Aires
17. Manila

Additionally, AWS plans to broaden Local Zones to encompass 19 additional locations spanning 16 countries:

1. Australia
2. Austria
3. Belgium
4. Brazil
5. Canada
6. Colombia
7. Czech Republic
8. Germany
9. Greece
10. India
11. Kenya
12. Netherlands
13. Norway
14. Portugal
15. South Africa
16. Vietnam

6. AWS Edge Locations:

- **Definition:**

AWS Edge Locations are **endpoints for Amazon CloudFront**, strategically located around the world.

- **Function:**

They act as **caching servers**, storing copies of **frequently accessed content** like images, videos, web pages, and other static files **closer to end users**.

- **Performance Benefits:**

Requests are served from the **nearest Edge Location**, which **reduces latency** and **improves application or website performance**.

- **Content Retrieval:**

If content is not already cached at an Edge Location, it is **fetched from the origin server** (e.g., Amazon S3 or a custom origin) and **cached for future requests**.

7. AWS Regional Edge Caches:

- **Definition:**

Regional Edge Caches are **CloudFront sites** located strategically worldwide, **closer to viewers**.

- **Function:**

They deliver content **between the origin server and global Edge Locations**, improving content delivery efficiency.

- **Use Cases:**

Ideal for **less popular or dynamic content**, such as **user-generated content**, **e-commerce assets**, **news**, and **rapidly trending content**.

- **Benefit:**

Helps **reduce latency** and ensures **faster content delivery** to end users.

8. AWS Wavelength Zones

- AWS Wavelength brings **cloud services to the edge of 5G networks**.
- It supports **real-time gaming, AR/VR, video streaming**, and other applications needing **ultra-low latency and high bandwidth**.
- A Wavelength Zone is a **specialized infrastructure** within a **telecom provider's data center**.
- These zones are located **close to 5G base stations** for a **direct connection** between AWS resources and the 5G network.
- Proximity to 5G networks **reduces latency**, enabling **instant responses** for applications.
- Developers can run workloads on **AWS infrastructure at the 5G network edge**, combining cloud capabilities with **fast, low-latency 5G connectivity**.

9. Latest Scale (2025)

- 38 Regions
- 120 Availability Zones
- 400+ Edge Locations

10. Role in Cloud Computing

- Ensures redundancy, resilience, and compliance.
- Regions and AZs enable disaster recovery, data sovereignty, and high availability.



AWS Shared Responsibility Model

1. Principle

- The **AWS Shared Responsibility Model** defines the **division of security and compliance responsibilities** between AWS and its customers.
- **AWS Responsibility:** Security of the cloud, including **infrastructure, hardware, software, networking, and facilities** that run AWS services.
- **Customer Responsibility:** Security in the cloud, including **data, applications, and configurations** within their own AWS environment.

2. AWS Shared Responsibility Model – Examples and Implications

- **Examples:**

- AWS secures **physical infrastructure**, including data centers, servers, and facilities.
- Customers secure **operating systems, applications, and data**, including **firewalls, encryption, identity management, and compliance controls**.
- For **Amazon EC2 (IaaS)**: Customers manage **OS, applications, and firewall (security group) configurations**.
- For **Amazon S3**: Customers manage **data access policies and encryption methods**.

- **Implications:**

- Customers can **focus on securing their workloads**, while AWS handles the **platform's foundational security**.
- **Responsibilities vary** depending on the AWS service used (**IaaS vs. abstracted services**).
- Customers must **actively manage risks** and **cannot rely solely on AWS** for security.

3. What Is AWS Responsible For?

- AWS is responsible for **security of the cloud**, meaning it **protects and secures the infrastructure** that runs all AWS services.
- AWS responsibilities include:
 - **Protecting and securing AWS Regions, Availability Zones, and data centers**, including **physical security of buildings**.
 - **Managing hardware, software, and networking components** that run AWS services, such as **physical servers, host operating systems, virtualization layers, and networking components**.
- The **level of AWS responsibility** varies depending on the service type. AWS classifies services into **three categories**, each with specific security responsibilities.

Category	Examples of AWS Services in the Category	AWS Responsibility
Infrastructure services	Compute services, such as Amazon Elastic Compute Cloud (Amazon EC2)	AWS manages the underlying infrastructure and foundation services.
Container services	Services that require less management from the customer, such as Amazon Relational Database Service (Amazon RDS)	AWS manages the underlying infrastructure and foundation services, operating system, and application platform.
Abstracted services	Services that require very little management from the customer, such as Amazon Simple Storage Service (Amazon S3)	AWS operates the infrastructure layer, operating system, and platforms, as well as server-side encryption and data protection.

4. What Is the Customer Responsible For?

- Customers are responsible for **security in the cloud**, meaning they must **properly configure services and applications** and **ensure their data is secure**.
- The **level of responsibility** depends on the type of AWS service used:
 - Some services require customers to handle **all security configuration and management tasks**.
 - More abstracted services require customers to **manage only data and access controls**.
- Using the **three AWS service categories**, customers can determine their **specific security responsibilities** for each service they use.

Category	AWS Responsibility	Customer Responsibility
Infrastructure services	AWS manages the infrastructure and foundation services.	You control the operating system and application platform, as well as encrypting, protecting, and managing customer data.
Container services	AWS manages the infrastructure and foundation services, operating system, and application platform.	You are responsible for customer data, encrypting that data, and protecting it through network firewalls and backups.
Abstracted services	AWS operates the infrastructure layer, operating system, and platforms, as well as server-side encryption and data protection.	You are responsible for managing customer data and protecting it through client-side encryption.

- **Service Selection:** Consider **which AWS service** to use and review the **level of responsibility** required to secure it.
- **Compliance Alignment:** Ensure the **shared security model** aligns with your **IT security standards** and **applicable laws and regulations**.
- **Data Control:** Customers maintain **full control of their data** and are responsible for **security related to their content**.
- **Examples of Responsibilities:**
 - **Choosing an AWS Region** in accordance with **data sovereignty regulations**.
 - **Implementing data protection** mechanisms, such as **encryption and backups**.
 - **Using access control** to limit **who can access your data and AWS resources**.

Customer



Responsibility for
security 'in' the cloud



Customer Data



Platform, Applications, IAM



Operating System, Network & Firewall Configuration



Client-Side Data, Encryption
& Data Integrity Authentication



Server-Side Encryption



Networking Traffic Protection

AWS



Responsibility for
security 'of' the cloud



Software



Compute



Storage



Database



Networking



Hardware / AWS Global Infrastructure



Regions



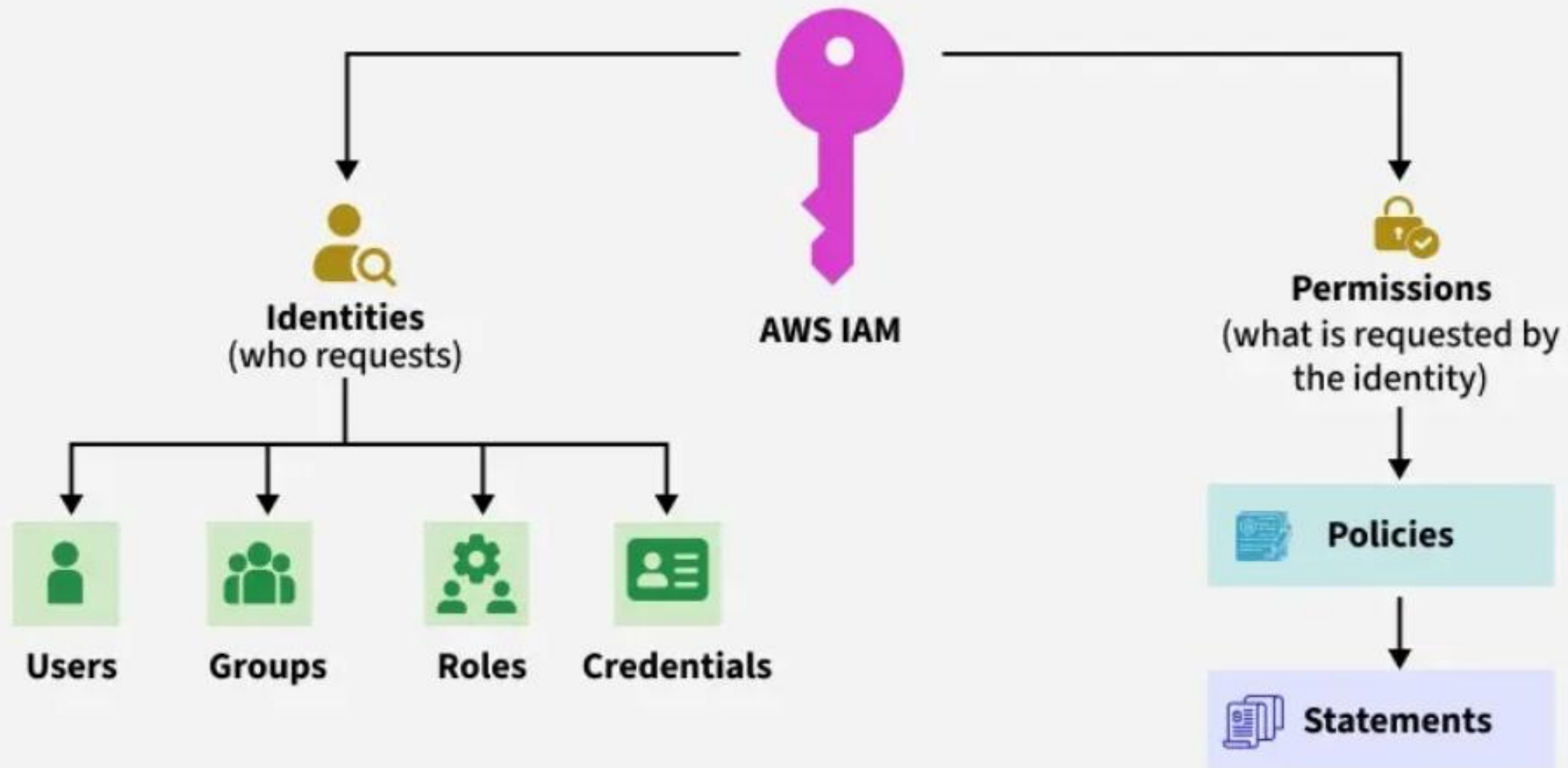
Availability Zones



Edge Locations

AWS Identity and Access Management (IAM)

- **AWS IAM** is a security service in AWS.
- It allows you to **control access** to AWS resources securely.
- You can **create and manage**:
 - Users
 - Groups
 - Roles
 - Permissions
- Ensures that **only authorized identities** can perform specific actions in AWS.



I. Use Cases of IAM

1. User and Group Management:

- Create individual users (e.g., developers, admins).
- Organize users into groups (e.g., DevTeam, Admins).
- Assign permissions based on roles or responsibilities.

• Example:

- Create a group **DevTeam** with **read-only access** to S3 buckets.
- All developers in the group **automatically inherit** these permissions.

2. Enhanced Security (Least Privilege Access)

- **Principle of Least Privilege:**

- Users get **only the permissions they need** for their tasks.
- Reduces risk of misconfiguration or misuse.

- **Example:**

- An **intern** in the DevOps team can **start/stop EC2 instances**.
- The intern **cannot terminate** instances, keeping critical infrastructure safe.

3. Access Management for Resources

- **Fine-Grained Control:**

- Decide **who can view, create, edit, or delete** specific resources.

- **Example:**

- A **project manager** can **view CloudWatch dashboards**.
 - Cannot **modify EC2 or RDS resources**, keeping control secure.

4. Multi-Factor Authentication (MFA) for Extra Security

- **Extra Protection:**

- Users must provide a **time-based one-time password (OTP)** along with their password.

- **Example:**

- The **root user account** is protected by MFA.
- Even if the password is stolen, **access is blocked without the OTP.**

II. Working of IAM

- **Authorization Check:** Verifies if a user or service is allowed to access a resource.
- **Identity Verification:** Checks **who is making the request**.
- **Permission Evaluation:** Looks at attached **permissions/policies** to decide if the action is allowed.



AWS Identity and Access Management

Apply Fine-Grained Permissions to AWS Services and Resources



Who
Workforce Users and Workloads with IAM



Can Access
Permissions with IAM Policies



What
Resources within your AWS Organization

Example: EC2 Access to S3

- **Goal:** Allow an EC2 instance to **upload files** to a specific S3 bucket.
- **Steps:**
 1. **Create an IAM Role** with `s3:GetObject` and `s3:PutObject` permissions.
 2. **Attach the role** to the EC2 instance.
 3. **IAM evaluates requests** and allows access if permissions match.

IAM Policy Snippet:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:GetObject", "s3:PutObject"],
      "Resource": "arn:aws:s3:::my-team-uploads/*"
    }
  ]
}
```

This example shows how IAM enables secure, role-based, and limited access, just like the "Who → Can Access → What" flow in the image.

Core Components of IAM

a. IAM Identities – Define **who can access AWS** and **what they can do**.

1. Root User

- Default user created when an AWS account is set up.
- Has **full administrative access**.
- Should be **protected with MFA**.
- Use separate **admin-level IAM users** for daily operations.

2. IAM Users

- Individual accounts for **people or services** needing AWS access.
- **Permissions assigned via policies**.

Example for IAM Users

- **User user-1:**
 - Given **read-only access** to EC2.
 - Cannot **create, modify, or delete instances**.



Users



Groups

**Core Components of AWS Identity
and Access Management (IAM)**



Roles



Policies

IAM enables centralized identity and access control in AWS.

3. IAM Groups

- A **group** is a collection of IAM users.
- **Simplifies access management** by assigning permissions to the group instead of individual users.

Example:

- **Group:** AutoScalingAdmins
- **user-1:** Can **modify Auto Scaling groups only**
- **user-2:** Has **broader permissions**, including EC2 management

4. IAM Roles

- Provide **temporary permissions** for AWS services or users.
- Commonly used for **service-to-service communication**.

Example:

- **Amazon EKS** needs to scale EC2 instances.
- Create a **role** with EC2 permissions.
- Attach the **role to the EKS cluster**.

5. IAM Policies

- **Define permissions:** what actions identities can perform on which resources.
- **Written in JSON** and include:
 - **Effect:** `Allow` or `Deny`
 - **Action:** AWS service operations (e.g., `s3:GetObject`)
 - **Resource:** Specific resources (e.g., an S3 bucket)